

SDE II Full-Stack Assessment

The objective is to evaluate their frontend, backend, database design, API design, code quality, and end-to-end product development skills.

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Video Learning + Interactive Quiz Builder

Objective

Build a mini learning platform where admins can create video-based learning content and add interactive questions that appear at specific timestamps during video playback.

Roles

Admin

- Create, update, and manage videos
- Add title, description, thumbnail, and video URL
- Add timestamp-based questions
- Publish or unpublish videos
- Assign videos to learners
- View learner progress and responses

Learner

- View assigned videos
- Watch video content
- Answer questions during playback
- Resume from the last watched position
- Complete the assigned video

Core Requirements

- JWT-based authentication
- Role-based access for Admin and Learner
- Video CRUD and assignment management
- Question types:
 - Single Choice
 - Multiple Choice
 - Short Answer
- Pause the video when a timestamp question appears
- Save learner responses
- Track:
 - Last watched timestamp
 - Completion percentage
 - Completion status
- Admin reporting for learner progress and responses

Expectations

- Clean and responsive React frontend
- Reusable components
- Proper loading, error, and empty states
- Structured Node.js/NestJS backend
- Proper validations and authorization
- Suitable MongoDB schema design
- Clean API structure and error handling
- Resume and progress tracking should work after refresh

Good-to-Have

- Progress autosave
- Search and filters
- Unit tests
- Docker-based local setup

Evaluation

Area	Weight
Frontend Implementation	30%
Backend APIs	25%
State Management	15%
Database Design	10%
User Experience	10%
Code Quality and Documentation	10%

Submission Requirements

Submit the following:

- Complete frontend and backend source code
- Git repository link
- Meaningful commit history
- README containing:
 - Local setup instructions
 - Environment variables
 - Database setup
 - Seed data instructions
 - Sample login credentials
 - Assumptions and known limitations
- API documentation using Swagger, Postman, or equivalent
- Working Loom video demonstrating:
 - Admin flow
 - Learner or candidate flow
 - Frontend and backend integration
 - Authentication and authorization
 - Major features
 - Important validations and edge cases

- Brief architecture explanation

Deployment

Deployment is **not required**.

The project must run locally using the instructions provided in the README.

Loom Video

Recommended duration: **8-12 minutes**

Expected Completion Time

8-12 hours

Recommended Stack

- React.js
- Node.js / NestJS
- MongoDB
- JWT Authentication